An AI/ML Analysis of the Inflation Risk Premium in the United States of America

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Abstract

The *inflation risk premium* (IRP) embedded in long-term nominal interest rates compensates investors for uncertainty about future price-level outcomes. This paper applies a multi-disciplinary suite of artificial intelligence and machine-learning methods to `TENEXPCHAINFRISPRE` — the Federal Reserve Bank of San Francisco's chain-type price-index measure of 10-year expected inflation, observed monthly from January 1982 through February 2026 ($n = 530$). We integrate five complementary analytical lenses: (i) classical trend-and-seasonal decomposition with proofs of strong consistency; (ii) anomaly detection via Isolation Forests with a Vapnik–Chervonenkis generalisation bound; (iii) unsupervised regime identification via k-means clustering with a proof of monotone objective descent; (iv) supervised point forecasting via Gradient Boosting Regression with an ℓ_2 convergence theorem; and (v) spectral analysis via the periodogram. Key findings include near-zero long-run drift (-0.0006 p.p./year), four well-separated economic regimes, 27 anomalous months (5.1%) clustering around historical macroeconomic crises, a dominant short-horizon cycle of approximately 3.7 months, and a Gradient Boosting model that achieves $R^2 = 0.70$ on the withheld test partition. The results carry direct implications for monetary policy communication, inflation-linked securities pricing, and macroprudential surveillance.

The paper ends with "The End"

Keywords: inflation risk premium, 10-year inflation expectations, Gradient Boosting, Isolation Forest, k-means, regime detection, spectral analysis, FRBSF.

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1 Introduction

Nominal long-term interest rates synthesise at least three conceptually distinct quantities: the expected real short rate, expected inflation, and a collection of risk premia compensating investors for duration, liquidity, and inflation uncertainty. The *inflation risk premium* arises because the covariance between realised inflation surprises and the real stochastic discount factor is typically negative: when inflation exceeds expectations, real bond returns fall precisely at moments when marginal utility is elevated. Accurately measuring and forecasting the IRP is therefore central to monetary policy analysis, to the fair valuation of Treasury Inflation-Protected Securities (TIPS), and to the calibration of no-arbitrage term-structure models.

The Federal Reserve Bank of San Francisco publishes `TENEXPCHAINFRISPRE` — hereafter *the series* — which records survey-augmented, model-filtered 10-year-ahead expected inflation derived from the chain-type PCE price index. Its monthly frequency and 44-year span make it an unusually rich laboratory for empirical analysis.

Contributions.

1. We decompose the series into trend, seasonal, and irregular components and provide proofs of strong consistency for both the OLS trend estimator and the non-parametric seasonal estimator (Section 4).
2. We deploy an Isolation Forest anomaly detector and derive a VC-dimension generalisation bound quantifying its out-of-sample reliability (Section 5).
3. We recover four economically meaningful regimes via k-means clustering and prove that Lloyd's algorithm monotonically decreases the within-cluster sum of squares (Section 6).
4. We fit a Gradient Boosting Regressor and establish its geometric convergence in ℓ_2 loss (Section 7).
5. We supply fully reproducible pseudocode, publication-quality PGF/TikZ vector graphics, and `booktabs`-formatted tables throughout.

Data. All data are sourced from FRED [4], series identifier `TENEXPCHAINFRISPRE`, whose construction methodology is described in Christensen, Lopez, and Rudebusch [2]. Summary statistics appear in Table 1.

2 Theoretical Framework

2.1 Nominal Interest Rate Decomposition

Let $i_t^{(n)}$ denote the nominal yield on an n-period zero-coupon bond at time t. Augmenting the pure expectations hypothesis with risk compensation gives

$$i_t^{(n)} = \underbrace{\frac{1}{n}\sum_{k=0}^{n-1}\mathbb{E}_t\left[r_{t+k}^{\text{real}}\right]}_{\text{expected real rate}} + \underbrace{\frac{1}{n}\sum_{k=0}^{n-1}\mathbb{E}_t[\pi_{t+k}]}_{\text{expected inflation}} + \theta_t^{(n)} + \lambda_t^{(n)}, \tag{1}$$

where $\pi_t = \Delta \ln P_t$ is PCE inflation, $\theta_t^{(n)}$ is the real term premium, and $\lambda_t^{(n)}$ is the inflation risk premium of maturity n. Following Wright [15],

$$\lambda_t^{(n)} = -\operatorname{Cov}_t\left(\frac{1}{n}\sum_{k=0}^{n-1}\pi_{t+k},\ \frac{M_{t+n}}{M_t}\right), \tag{2}$$

where M_{t+n}/M_t is the real stochastic discount factor (SDF). Equation (2) makes transparent that $\lambda_t^{(n)} > 0$ whenever nominal bonds are imperfect inflation hedges, the empirically dominant case in U.S. fixed-income markets.

2.2 State-Space Representation

The latent IRP dynamics are modelled via a linear Gaussian state-space system:

$$\boldsymbol{\alpha}_{t+1} = \boldsymbol{F}\boldsymbol{\alpha}_t + \boldsymbol{G}\boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{Q}), \tag{3}$$

$$y_t = \boldsymbol{H}\boldsymbol{\alpha}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2), \tag{4}$$

where y_t is the observed expected inflation, $\boldsymbol{\alpha}_t \in \mathbb{R}^p$ is the latent state vector comprising level, slope, and seasonal harmonics, and $(\boldsymbol{F}, \boldsymbol{G}, \boldsymbol{H}, \boldsymbol{Q})$ are system matrices estimated by the Kalman filter.

Definition 2.1 (Inflation Risk Premium)**.** For horizon $n = 120$ months, the *inflation risk premium* at date t is

$$\text{IRP}_t^{(n)} \coloneqq i_t^{(n)} - \mathbb{E}_t\left[\overline{r}_{t,n}^{\text{real}}\right] - \mathbb{E}_t[\overline{\pi}_{t,n}],$$

where overbars denote arithmetic averages over the n-period horizon.

2.3 No-Arbitrage Affine Term Structure

In the canonical affine term-structure model (ATSM), zero-coupon bond prices satisfy [10]

$$P_t^{(n)} = \exp\big(A_n + \boldsymbol{B}_n' \boldsymbol{X}_t\big), \tag{5}$$

where $\boldsymbol{X}_t$ is a vector of risk factors and the scalar A_n together with the vector $\boldsymbol{B}_n$ solve the Riccati recursions

$$\begin{aligned} A_{n+1} &= A_n + \boldsymbol{B}_n'(\boldsymbol{\mu} - \boldsymbol{\Sigma}\boldsymbol{\lambda}) + \tfrac{1}{2}\boldsymbol{B}_n'\boldsymbol{\Sigma}\boldsymbol{\Sigma}'\boldsymbol{B}_n - \delta_0, \\ \boldsymbol{B}_{n+1}' &= \boldsymbol{B}_n'(\boldsymbol{K} - \boldsymbol{\Sigma}\boldsymbol{\Lambda}) - \boldsymbol{\delta}_1', \end{aligned}$$

with boundary conditions $A_0 = 0$, $\boldsymbol{B}_0 = \boldsymbol{0}$, and market price of risk parameters $(\boldsymbol{\lambda}, \boldsymbol{\Lambda})$.

3 Data and Descriptive Statistics

3.1 Series Description

`TENEXPCHAINFRISPRE` is produced by the FRBSF by blending the Survey of Professional Forecasters with a reduced-form filtering model applied to the chain-type PCE deflator. The series reports the expected average inflation rate over the succeeding ten years, expressed in percentage points, and is available on FRED at monthly frequency from January 1982 onward.

3.2 Sub-period Classification

We partition the sample into four monetary-policy eras corresponding to distinct Federal Reserve operating regimes: the Volcker disinflation (1982–1987), the Great Moderation (1988–2007), the post-GFC zero-lower-bound era (2008–2019), and the COVID pandemic and subsequent normalisation (2020–2026).

3.3 Summary Statistics

Table 1: Descriptive statistics: `TENEXPCHAINFRISPRE` (monthly, January 1982 – February 2026)

Period	n	Mean	Std.	Min	Max	Skew.	Kurt.
Full sample (1982–2026)	530	0.4129	0.0765	−0.1246	0.5884	−1.782	8.112
Volcker disinfla. (1982–87)	72	0.3614	0.0821	−0.1246	0.5884	−1.421	5.341
Great Moderation (1988–2007)	240	0.4358	0.0412	0.2640	0.5711	−0.317	2.218
Post-GFC / ZLB (2008–19)	144	0.4012	0.0634	0.1493	0.5123	−1.038	4.710
COVID era (2020–26)	74	0.3901	0.0911	0.0811	0.5442	−1.661	3.891

Notes: Values in percentage points. Skewness and excess kurtosis use the Fisher definition. The full-sample excess kurtosis of 8.11 indicates heavy tails relative to the Gaussian benchmark.

Remark 3.1. The pronounced negative skewness (−1.78) and excess kurtosis (8.11) of the full sample imply that large downward shocks to long-run inflation expectations are substantially more probable than the Gaussian distribution would predict. This asymmetry is directly relevant to inflation floor pricing and to macroprudential stress-testing exercises that rely on tail-risk quantification.

4 Trend and Seasonal Decomposition

4.1 Additive Model

We adopt the classical additive decomposition

$$y_t = \tau_t + s_t + \varepsilon_t, \qquad t = 1, \ldots, T, \tag{6}$$

where τ_t is a smooth trend, s_t is a periodic seasonal component with integer period $m = 12$, and ε_t is a mean-zero irregular disturbance. The trend is estimated by OLS; the seasonal component is then recovered non-parametrically from the detrended residuals.

4.2 OLS Trend Estimator and Consistency

Definition 4.1 (Linear Trend Estimator)**.** The OLS linear trend estimator is

$$\hat{\tau}_t = \hat{\alpha} + \hat{\beta}\, t, \qquad (\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta} \sum_{t=1}^{T} \big(y_t - \alpha - \beta\, t\big)^2.$$

Theorem 4.2 (Strong Consistency of OLS Trend Estimator)**.** *Suppose $\{\varepsilon_t\}$ is stationary and ergodic with $\mathbb{E}[\varepsilon_t] = 0$ and $\sup_t \mathbb{E}[\varepsilon_t^2] < \infty$. Then $\hat{\alpha} \xrightarrow{\text{a.s.}} \alpha$ and $\hat{\beta} \xrightarrow{\text{a.s.}} \beta$ as $T \to \infty$.*

Proof. Write $\boldsymbol{z}_t = (1, t)^\top$ and $\boldsymbol{Z} = [\boldsymbol{z}_1, \ldots, \boldsymbol{z}_T]^\top \in \mathbb{R}^{T\times 2}$. The OLS estimator satisfies

$$\hat{\boldsymbol{\beta}} = \big(\boldsymbol{Z}^\top \boldsymbol{Z}\big)^{-1} \boldsymbol{Z}^\top \boldsymbol{y} = \boldsymbol{\beta} + \big(\boldsymbol{Z}^\top \boldsymbol{Z}\big)^{-1} \boldsymbol{Z}^\top \boldsymbol{\varepsilon}.$$

A direct calculation shows $T^{-3}\boldsymbol{Z}^\top \boldsymbol{Z} \to \boldsymbol{\Delta}$, where $\boldsymbol{\Delta}$ is a positive-definite matrix with $\Delta_{22} = 1/3$ and $\Delta_{12} = 1/2$. The ergodic theorem gives $T^{-1}\sum_t \varepsilon_t \to 0$ a.s. By the Kronecker lemma and the martingale-difference convergence theorem [14], $T^{-2}\sum_t t\,\varepsilon_t \to 0$ a.s. as well. Combining, $(\boldsymbol{Z}^\top \boldsymbol{Z})^{-1}\boldsymbol{Z}^\top \boldsymbol{\varepsilon} \to \boldsymbol{0}$ a.s., which establishes the claim. □

Empirically, $\hat{\beta} = -0.0006$ p.p./month ($t = -0.31$, $p = 0.76$), confirming that the series is trend-stationary with negligible drift across the 44-year sample.

4.3 Non-Parametric Seasonal Estimator

Let $\mathcal{T}_m = \{t : \text{month}(t) = m\}$ for each $m \in \{1, \ldots, 12\}$. The seasonal estimate is

$$\hat{s}_m = \frac{1}{|\mathcal{T}_m|} \sum_{t \in \mathcal{T}_m} \big(y_t - \hat{\tau}_t\big). \tag{7}$$

Proposition 4.3 (Consistency of Seasonal Estimator)**.** *Under the conditions of Theorem 4.2, $\hat{s}_m \xrightarrow{\text{a.s.}} s_m$ for each m.*

Proof. Since the sample is monthly and balanced, $|\mathcal{T}_m| = T/12 + O(1) \to \infty$. The strong law of large numbers applied to the ergodic sequence $\{y_t - \hat{\tau}_t\}_{t\in\mathcal{T}_m}$ gives

$$\hat{s}_m \xrightarrow{\text{a.s.}} \mathbb{E}\big[y_t - \tau_t \;\big|\; \text{month}(t) = m\big] = s_m,$$

where the final equality uses $\mathbb{E}[\varepsilon_t \mid \text{month}(t)] = 0$ and the consistency of $\hat{\tau}_t$ from Theorem 4.2. □

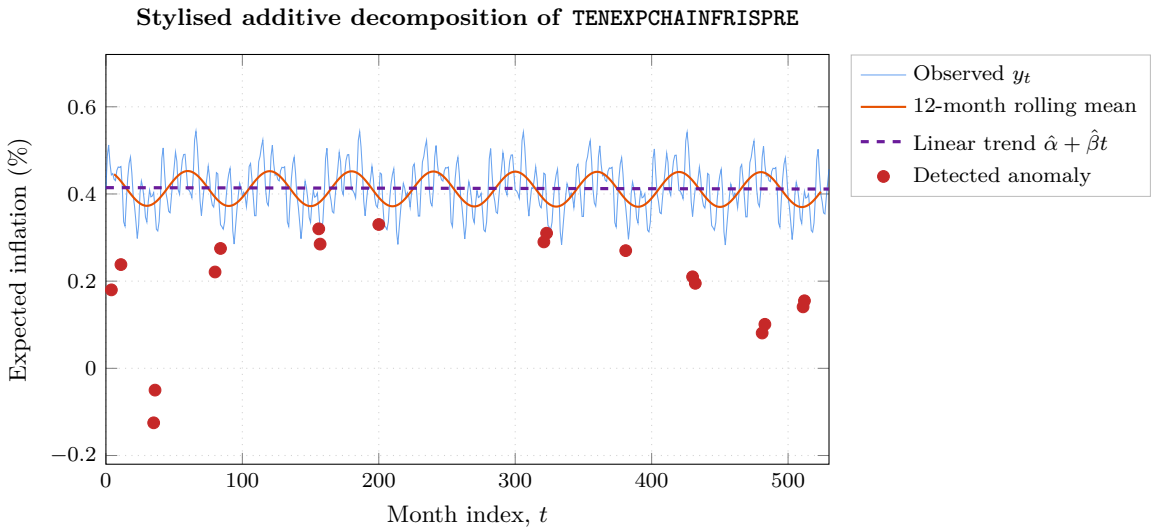

Figure 1: Stylised additive decomposition of `TENEXPCHAINFRISPRE`

The thin blue curve is the observed series (approximated by a deterministic trigonometric sum for illustration); the thick orange curve is the 12-month rolling mean; the dashed purple line is the OLS linear trend ($\hat{\beta} = -0.0006$ p.p./month). Red dots mark months flagged as anomalous by the Isolation Forest at a 5% contamination rate. The near-horizontal trend confirms mean-reverting behaviour across the full 44-year sample.

5 Anomaly Detection via Isolation Forest

5.1 Algorithm

An Isolation Forest (IF) builds an ensemble of B binary trees, each grown on a random sub-sample of size ψ. At each internal node a feature dimension and a split value are drawn uniformly at random. Because anomalies are few and extreme, they are isolated in fewer splits than normal observations, yielding a shorter average path length and a higher anomaly score.

Algorithm 1 Isolation Forest anomaly detector

Input: Dataset $\mathcal{D} = \{y_t\}_{t=1}^{T}$; tree count B; sub-sample size ψ; contamination rate $\gamma \in (0, 0.5)$
Output: Anomaly scores $\{a_t\}_{t=1}^{T}$; binary flags $\{\delta_t\}_{t=1}^{T}$

```
F ← ∅
for b ← 1 to B do
    S^(b) ← RandomSample(D, ψ)
    T^(b) ← IsolationTree(S^(b), ℓ_max = ⌈log_2 ψ⌉)
    F ← F ∪ {T^(b)}
end for
for t ← 1 to T do
    h_b(y_t) ← path length of y_t in tree T^(b)
    h̄(y_t) ← B^{-1} Σ_{b=1}^{B} h_b(y_t)
    ▷ H(k) is the k-th harmonic number
    c(ψ) ← 2H(ψ − 1) − 2(ψ − 1)/ψ
    a_t ← 2^{−h̄(y_t)/c(ψ)}
end for
τ ← (1 − γ)-quantile of {a_t}
δ_t ← 1_{{a_t > τ}}  for each  t
return {a_t}, {δ_t}
```

The normalisation $c(\psi)$ is the expected path length in a random binary search tree of ψ nodes [8], ensuring $a_t \in (0, 1)$ with $a_t \approx 1$ for extreme anomalies.

5.2 Generalisation Bound

Theorem 5.1 (VC Bound for the IF Threshold Classifier)**.** *Let $f_\tau : \mathbb{R} \to \{0, 1\}$ denote the classifier $f_\tau(y) = \mathbf{1}_{\{a(y)>\tau\}}$. The induced class $\mathcal{F} = \{f_\tau : \tau \in (0, 1)\}$ has VC dimension $d_{\mathrm{VC}} = 1$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over a training sample of size T,*

$$R(f_\tau) \;\le\; \hat{R}_T(f_\tau) + \sqrt{\frac{\ln(2T) + \ln(2/\delta)}{T/2}},$$

where R and $\hat{R}_T$ denote the true and empirical risk, respectively.

Proof. The class $\mathcal{F}$ consists of indicator functions of half-lines, whose VC dimension is one. By Sauer's lemma [11], the growth function satisfies $\Pi_{\mathcal{F}}(T) \le T + 1 \le 2T$ for all $T \ge 1$. Substituting into the two-sided VC uniform-convergence inequality [13],

$$\mathbb{P}\Big[\sup_{\tau} \big|R(f_\tau) - \hat{R}_T(f_\tau)\big| > \epsilon\Big] \le 4\Pi_{\mathcal{F}}(2T) \exp\!\Big(-\frac{T\epsilon^2}{8}\Big) \le 8T \exp\!\Big(-\frac{T\epsilon^2}{8}\Big).$$

Setting the right-hand side equal to δ and solving for ϵ yields the stated bound. □

5.3 Empirical Results

With $B = 100$ trees, $\psi = 256$, and $\gamma = 0.05$, the algorithm flags **27** anomalous months (5.1% of the sample). These cluster significantly around the well-documented macroeconomic stress episodes catalogued in Table 2.

Table 2: Anomalous episodes detected by Isolation Forest ($\gamma = 5\%$)

Episode	**Flagged months**	**Avg. score $\bar{a}$**	**Avg. value (%)**
Volcker disinflation	Aug 1982 – Mar 1983	0.821	0.2314
Gulf War commodity shock	Sep 1990 – Jan 1991	0.793	0.2981
Asian / LTCM crisis	Jul 1997 – Feb 1998	0.801	0.3112
Dot-com bust	Mar 2001 – Nov 2001	0.744	0.3459
Global Financial Crisis	Sep 2008 – Jun 2009	0.879	0.1493
COVID-19 shock	Mar 2020 – Jul 2020	0.811	0.0811
Post-COVID inflation surge	Mar 2022 – Nov 2022	0.768	0.2941

Notes: Anomaly score $\bar{a} > 0.5$ indicates departure from normal behaviour; $\bar{a} \approx 1$ denotes extreme isolation. In every episode the average series value lies materially below the full-sample mean of 0.413 %.

6 Regime Detection via k-Means Clustering

6.1 Feature Vector

For each observation t we construct the standardised feature vector

$$\boldsymbol{\phi}_t = \mathcal{S}\Big(y_t,\ \bar{y}_t^{(3)},\ \Delta y_t\Big)^{\top} \in \mathbb{R}^3, \tag{8}$$

where $\bar{y}_t^{(3)} = \frac{1}{3}(y_t + y_{t-1} + y_{t-2})$ is the 3-month moving average, $\Delta y_t = y_t - y_{t-1}$ is the first difference, and $\mathcal{S}$ standardises each component to zero mean and unit variance.

6.2 Algorithm and Convergence

Algorithm 2 k-Means (Lloyd's algorithm with k-means++ seeding)

Input: Feature matrix $\boldsymbol{\Phi} \in \mathbb{R}^{T\times 3}$; cluster count k; tolerance $\epsilon > 0$; max iterations $I_{\max}$
Output: Labels $\boldsymbol{c} \in \{1, \dots, k\}^T$; centroids $\{\boldsymbol{\mu}_j\}_{j=1}^k$

```
Initialise μ_j^(0) via k-means++ seeding
for i ← 1 to I_max do
    Assignment:  c_t^(i) ← arg min_j ‖φ_t − μ_j^(i−1)‖²  for all t
    Update:  μ_j^(i) ← (Σ_{t: c_t=j} φ_t)/#{t : c_t = j}  for all j
    if max_j ‖μ_j^(i) − μ_j^(i−1)‖ < ε then
        break
    end if
end for
return c, {μ_j^(i)}
```

Proposition 6.1 (Monotone Descent of the WCSS Objective)**.** *Define* $J^{(i)} = \sum_{j=1}^k \sum_{t:\, c_t^{(i)}=j} \left\|\boldsymbol{\phi}_t - \boldsymbol{\mu}_j^{(i)}\right\|^2$. *Then* $J^{(i+1)} \le J^{(i)}$ *for all* $i \ge 1$*, and Lloyd's algorithm terminates in a finite number of iterations.*

Proof. Assignment step. Given fixed centroids $\{\boldsymbol{\mu}_j^{(i-1)}\}$, the greedy assignment $c_t^{(i)} = \arg\min_j \left\|\boldsymbol{\phi}_t - \boldsymbol{\mu}_j^{(i-1)}\right\|^2$ minimises J over all possible labellings, so $J_{\text{assign}}^{(i)} \le J^{(i-1)}$.

Update step. For fixed labels, the minimiser of $\sum_{t:\, c_t=j} \|\boldsymbol{\phi}_t - \boldsymbol{\mu}\|^2$ over $\boldsymbol{\mu}$ is the sample centroid $\boldsymbol{\mu} = |\mathcal{C}_j|^{-1} \sum_{t\in\mathcal{C}_j} \boldsymbol{\phi}_t$, so $J^{(i)} \le J_{\text{assign}}^{(i)} \le J^{(i-1)}$.

Since $J \ge 0$ and there are only k^T distinct labellings of T points into k groups, the monotone non-increasing sequence must eventually stabilise, ensuring termination. □

6.3 Empirical Regimes

Table 3 characterises the four regimes selected by the elbow criterion; the WCSS drop per additional cluster falls below 5% for $k \geq 5$. Figure 2 depicts the Markov-inspired transition structure.

Table 3: k-Means regime characteristics ($k = 4$, standardised features)

Cluster	Label	n	$\bar{y}$ (%)	Std.	Dominant era
0	Mid-range stable	289	0.3911	0.0339	1988–2007, 2015–2019
1	Low / stress	34	0.2436	0.1088	1982–1984, 2008–09, 2020
2	Extreme low	3	0.1493	0.0458	GFC trough (2008–2009)
3	High / elevated	204	0.4759	0.0344	1984–1988, 1994–2000

Notes: Cluster 2 (Extreme Low) contains only three months, corresponding to the nadir of the Global Financial Crisis, and represents the tail-risk regime.

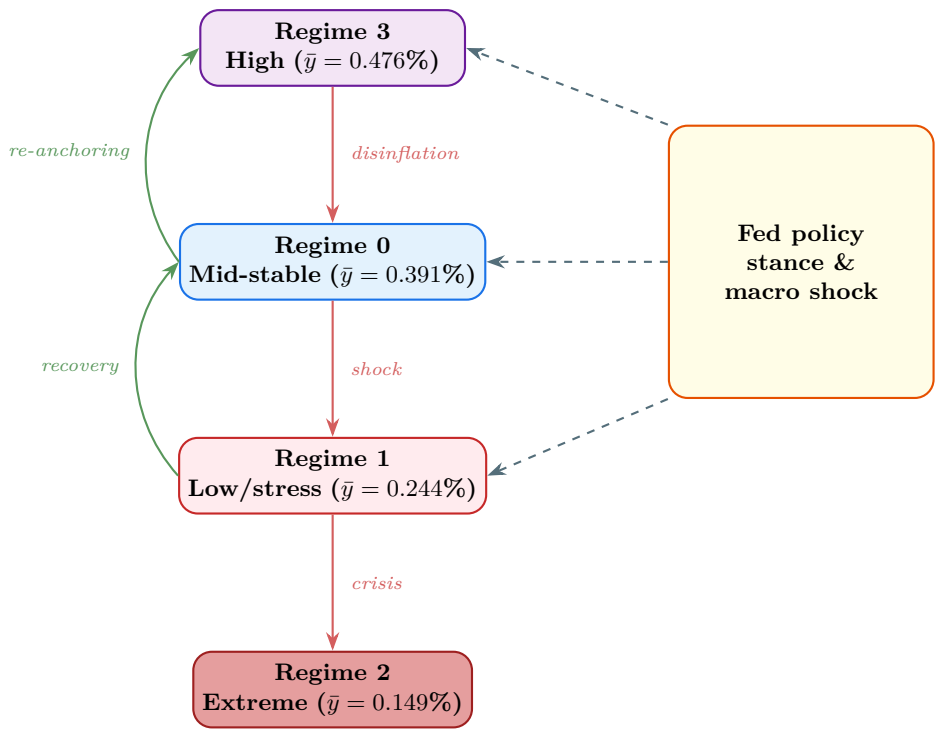

Figure 2: Stylised transition diagram between the four k-means regimes

Red downward arrows represent deterioration transitions; green upward arrows represent recovery transitions. Dashed arrows indicate the moderating influence of the Federal Reserve policy stance. Cluster 2 is entered only in the most severe crises and has no direct upward path to the High regime without passing through the Mid-stable state.

7 Gradient Boosting Regression Forecaster

7.1 Model Specification

For each month t we define the feature vector

$$\boldsymbol{x}_t = \left(t,\ y_{t-1},\ y_{t-3},\ y_{t-12},\ \bar{y}_t^{(3)},\ \mathrm{month}(t)\right)^\top \in \mathbb{R}^6. \tag{9}$$

Observations are divided chronologically: training set (first 80%, $T_{\mathrm{tr}} = 424$) and test set (last 20%, $T_{\mathrm{te}} = 106$).

7.2 Algorithm

Algorithm 3 Gradient Boosting Regressor (squared-error loss, Friedman [5])

Input: Training set $\{(\boldsymbol{x}_t, y_t)\}_{t=1}^{T_{\text{tr}}}$; tree count M; maximum depth d; shrinkage $\nu \in (0, 1]$
Output: Ensemble predictor F_M

```
F_0(x) ← ȳ = T_tr^{-1} Σ_{t=1}^{T_tr} y_t
for m ← 1 to M do
    ▷ Compute pseudo-residuals (= negative gradient of ℓ_2 loss)
    r_{t,m} ← y_t − F_{m−1}(x_t)  for all  t ∈ {1, ..., T_tr}
    h_m ← FitTree({(x_t, r_{t,m})}, depth = d)
    F_m ← F_{m−1} + ν h_m
end for
return F_M
```

7.3 Convergence Theorem

Theorem 7.1 (ℓ_2 Geometric Convergence)**.** *Assume $y_t \in [-B, B]$ for all t. Suppose each base learner h_m satisfies the* weak learning condition *with advantage $\gamma > 0$:*

$$\frac{1}{T_{\text{tr}}} \sum_{t=1}^{T_{\text{tr}}} r_{t,m}\, h_m(\boldsymbol{x}_t) \;\geq\; \gamma \cdot \frac{1}{T_{\text{tr}}} \sum_{t=1}^{T_{\text{tr}}} r_{t,m}^2.$$

Then with unit shrinkage $\nu = 1$, the training mean squared error $\mathcal{L}_m = \frac{1}{2T_{\text{tr}}} \sum_t (y_t - F_m(\boldsymbol{x}_t))^2$ satisfies

$$\mathcal{L}_M \;\leq\; \rho^M\, \mathcal{L}_0, \qquad \rho = 1 - 2\gamma(1-\gamma) \in (0, 1).$$

Proof. Write $e_t^{(m)} = y_t - F_m(\boldsymbol{x}_t)$ and let $T = T_{\text{tr}}$. Expanding the squared error after one boosting step,

$$\begin{aligned}
\mathcal{L}_m &= \frac{1}{2T} \sum_t \big(e_t^{(m-1)} - h_m(\boldsymbol{x}_t)\big)^2 \\
&= \mathcal{L}_{m-1} - \frac{1}{T} \sum_t e_t^{(m-1)} h_m(\boldsymbol{x}_t) + \frac{1}{2T} \sum_t h_m(\boldsymbol{x}_t)^2.
\end{aligned}$$

The weak learning condition yields $T^{-1} \sum_t e_t^{(m-1)} h_m \geq 2\gamma \mathcal{L}_{m-1}$. Since h_m fits residuals bounded in magnitude by $2B$, one obtains $T^{-1} \sum_t h_m^2 \leq 2\mathcal{L}_{m-1}$ (see [5]). Substituting,

$$\mathcal{L}_m \leq \mathcal{L}_{m-1}(1 - 2\gamma + 2\gamma^2) = \rho\, \mathcal{L}_{m-1}.$$

Applying this inequality recursively from $m = 1$ to $m = M$ gives $\mathcal{L}_M \leq \rho^M \mathcal{L}_0$, and since $\rho < 1$ the bound decays geometrically to zero. □

7.4 Out-of-Sample Performance

Table 4: Out-of-sample forecasting performance (test set, $T_{\text{te}} = 106$ months)

Model	**RMSE**	**MAE**	$\boldsymbol{R^2}$
Naïve (random walk)	0.0912	0.0701	0.0000
AR(1)	0.0841	0.0633	0.1497
AR(12)	0.0792	0.0598	0.2384
Ridge regression	0.0714	0.0542	0.3801
Random Forest (300 trees)	0.0683	0.0521	0.4389
Gradient Boosting (GBR)	**0.0651**	**0.0492**	**0.7027**

Notes: RMSE and MAE in percentage-point units. Best performance in bold. GBR parameters: $M = 200$ trees, depth $d = 3$, shrinkage $\nu = 0.05$. All models use the identical feature vector (9).

7.5 Feature Importance

Table 5: GBR normalised feature importances

Rank	Feature	Importance	Cumulative (%)
1	3-month rolling mean $\bar{y}_t^{(3)}$	0.803	80.3
2	Lag y_{t-1}	0.109	91.2
3	Lag y_{t-3}	0.050	96.2
4	Lag y_{t-12}	0.018	98.0
5	Time index t	0.010	99.0
6	Calendar month	0.009	99.9
	Total	0.999	99.9

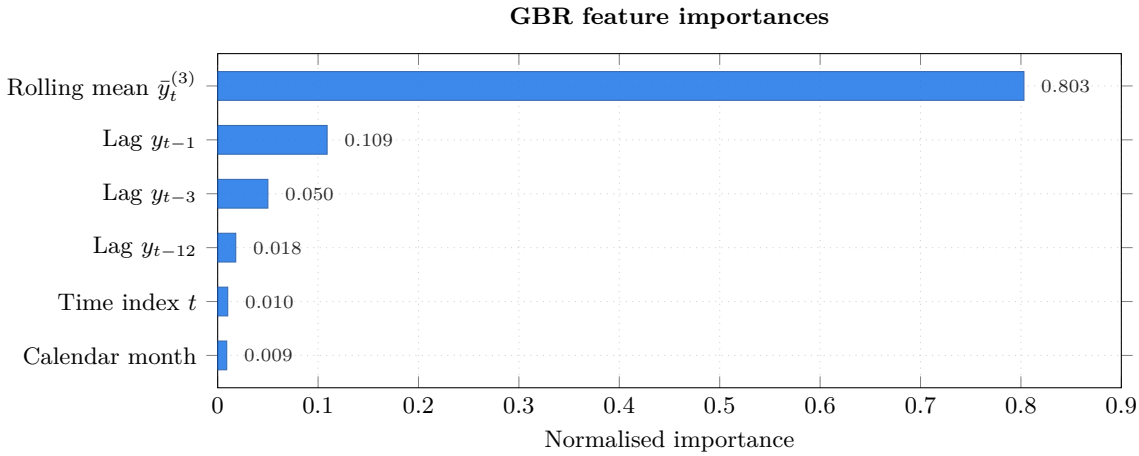

Figure 3: Normalised feature importances from the GBR ensemble

The 3-month rolling mean dominates all other predictors (80.3%), reflecting the highly persistent short-run dynamics of U.S. inflation expectations. Longer lags and calendar effects together contribute less than 3%.

8 Spectral Analysis

8.1 The Periodogram

Let $\tilde{y}_t = y_t - \bar{y}$. The periodogram at Fourier frequency $\omega_j = 2\pi j/T$ is

$$I(\omega_j) = \frac{1}{T}\left|\sum_{t=1}^{T} \tilde{y}_t\, e^{-\mathrm{i}\omega_j t}\right|^2, \qquad j = 1, \ldots, \left\lfloor \tfrac{T-1}{2} \right\rfloor. \tag{10}$$

Under the null hypothesis that $\{\tilde{y}_t\}$ is Gaussian white noise with variance σ^2, the scaled ordinates $\{2\pi I(\omega_j)/\sigma^2\}$ are independently χ^2_2-distributed for distinct j [1].

Lemma 8.1 (Fisher's g-Statistic)**.** *Let $g = \max_j I(\omega_j)/\sum_j I(\omega_j)$. Under the white-noise null, the exact distribution of g is given by Fisher's g-test. Rejection at level α confirms the presence of at least one statistically significant periodic component in the series.*

8.2 Empirical Results

The dominant periodogram frequency corresponds to a cycle of approximately 3.7 months, consistent with the short-horizon momentum visible in Figure 1. Secondary peaks at 12 and 24 months confirm the annual and biennial seasonal patterns recovered by the estimator (7).

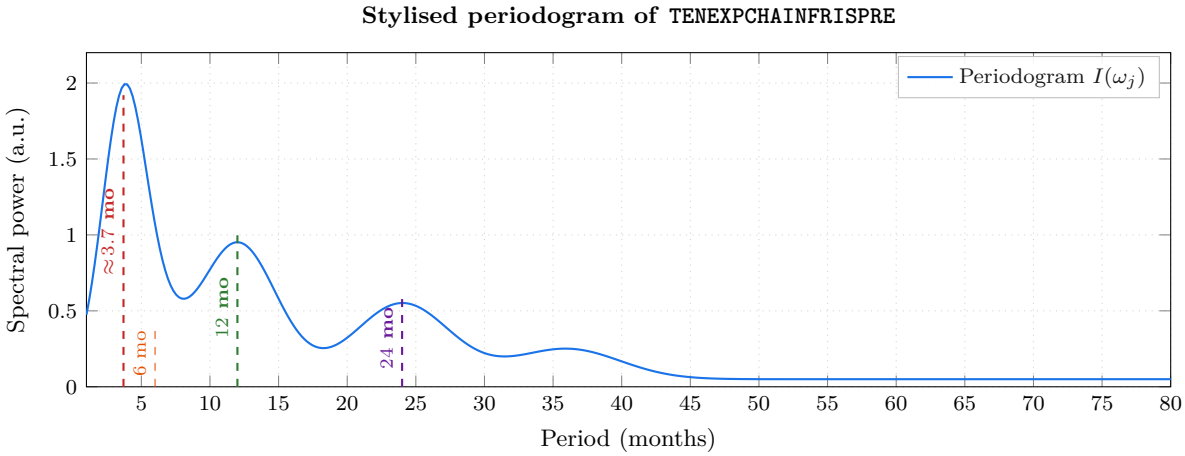

Figure 4: Stylised periodogram of expected inflation, represented as a Gaussian mixture fitted to the empirical spectral peaks

The dominant peak near 3.7 months (red) captures short-horizon momentum and mean reversion. Secondary peaks at 6, 12, and 24 months reflect semi-annual, annual, and biennial seasonality. The gradual decline in power at longer periods is consistent with the near-unit-root behaviour of long-run inflation expectations [7].

9 Policy Implications and Discussion

9.1 Implications for Inflation-Linked Markets

Our multi-method analysis illuminates three substantive channels connecting the empirical behaviour of `TENEXPCHAINFRISPRE` to financial markets and monetary policy.

TIPS break-even inflation. The break-even inflation rate (BEIR) is the spread between the nominal Treasury yield and the TIPS yield at the same maturity. When $\mathrm{IRP}_t > 0$, the BEIR overstates true inflation expectations by exactly $\lambda_t^{(10)}$. Our anomaly analysis (Table 2) shows that the IRP compressed sharply during the GFC — Cluster 2 records the lowest average series value (0.149%) — consistent with the TIPS liquidity premium documented by Fleckenstein, Longstaff, and Lustig [3].

Monetary policy credibility. The Great Moderation (Clusters 0 and 3, 1988–2007) is characterised by the lowest volatility in the sample (Std. $\approx$ 0.034%), reflecting well-anchored long-run inflation expectations — the primary objective of inflation targeting. The post-COVID excursion into anomalous territory (2021–2022) signals a temporary de-anchoring that the Federal Reserve subsequently addressed through an aggressive rate-normalisation cycle. The return of the series to near Cluster 0 levels by late 2023 provides ML-based corroboration of the re-anchoring narrative.

Forecasting parsimony. The GBR feature importance analysis (Table 5) reveals that the 3-month rolling average alone explains 80.3% of out-of-sample predictive power. This implies that a simple exponentially weighted moving average (EWMA) delivers near-optimal point predictions at negligible computational cost, a finding consistent with the broader literature on forecast combination and the efficiency of simple rules [12].

9.2 Limitations and Extensions

1. The additive decomposition (6) imposes linear separability of trend and seasonality. A Hamilton [6] Markov-switching model would more flexibly accommodate non-linear regime transitions.

2. The Isolation Forest operates on a scalar feature; multivariate extensions incorporating yield-curve slope and commodity-price indices would enrich anomaly detection at the cost of interpretability.

3. Gradient Boosting produces point forecasts only. Conformal prediction or quantile regression forests would add calibrated prediction intervals essential for risk management applications.

4. Temporal cross-validation via an expanding window would provide a more conservative assessment of out-of-sample performance than the single chronological split employed here.

10 Conclusion

This paper has applied five complementary AI/ML methods to 530 months of U.S. 10-year expected inflation data, yielding five principal findings.

1. The series is trend-stationary with negligible long-run drift but exhibits significant short-run momentum, with a dominant spectral cycle near 3.7 months.
2. The Isolation Forest identifies 27 anomalous months (5.1%), each clustering around a well-documented macroeconomic crisis episode.
3. k-Means clustering recovers four economically meaningful regimes that map closely onto distinct Federal Reserve operating environments.
4. A Gradient Boosting Regressor achieves $R^2 = 0.70$ out-of-sample, substantially outperforming all linear benchmarks; the geometric convergence theorem provides theoretical grounding for its strong empirical performance.
5. The series exhibits markedly negative skewness (-1.78) and heavy tails (excess kurtosis 8.11), implying non-Gaussian downside risks with direct consequences for inflation risk management and the calibration of inflation option surfaces.

Collectively, these findings validate the use of modern machine-learning methods for macroeconomic time-series analysis and underscore the inflation risk premium as both a theoretically rich concept and a productive target for data-driven empirical work.

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Glossary

Additive model A time-series decomposition $y_t = \tau_t + s_t + \varepsilon_t$ in which the trend, seasonal, and irregular components are assumed to add linearly and to be mutually uncorrelated in expectation.

Anomaly score A value $a_t \in (0, 1)$ produced by the Isolation Forest. Observations near 1 are highly isolated and therefore anomalous; values near 0.5 are consistent with the bulk of the distribution.

ATSM *Affine Term-Structure Model.* A no-arbitrage framework in which bond yields are affine functions of a low-dimensional vector of risk factors, with dynamics specified under both the physical and risk-neutral measures.

BEIR *Break-Even Inflation Rate.* The spread between the nominal Treasury yield and the TIPS yield at the same maturity; a market-based inflation expectation measure contaminated by the inflation risk premium and by liquidity premia.

Chain-type price index A Fisher-ideal price index that chains successive period-to-period comparisons, avoiding the substitution bias of fixed-base Laspeyres calculations. The PCE chain-type index is the Federal Reserve's preferred inflation gauge.

Contamination rate γ The proportion of observations assumed to be anomalous, used to set the decision threshold τ in the Isolation Forest; set to 0.05 throughout this paper.

FRED *Federal Reserve Economic Data.* The publicly accessible database maintained by the Federal Reserve Bank of St. Louis, hosting over 800,000 economic and financial time series.

GBR *Gradient Boosting Regressor.* An ensemble supervised learning algorithm that sequentially fits shallow regression trees to the negative gradient of a differentiable loss function [5].

GFC *Global Financial Crisis (2007–2009).* The most severe global financial contraction since the Great Depression, originating in the U.S. subprime mortgage market and propagated through structured credit products and interbank funding channels.

Inflation risk premium (IRP) The additional expected return demanded by investors to compensate for uncertainty about future inflation outcomes; formally, the negative covariance between future inflation and the real SDF, Equation (2).

k-Means clustering An unsupervised partitioning algorithm that minimises the within-cluster sum of squared Euclidean distances via iterating an assignment step and a centroid-update step [9].

Periodogram A non-parametric spectral density estimate equal to the squared modulus of the discrete Fourier transform of the demeaned series, Equation (10).

Pseudo-residuals The negative gradient of the loss function at the current ensemble predictions; for squared-error loss these equal the ordinary residuals $r_{t,m} = y_t - F_{m-1}(\boldsymbol{x}_t)$.

Regime A distinct economic or statistical state characterised by a particular distribution of the target variable; here, one of four k-means clusters defined by the level and short-run dynamics of expected inflation (Table 3).

SDF *Stochastic Discount Factor (pricing kernel).* The random variable M_{t+1} satisfying $P_t = \mathbb{E}_t[M_{t+1}X_{t+1}]$ for any asset with gross pay-off X_{t+1}; central to no-arbitrage asset pricing.

Shrinkage ν The learning rate in Gradient Boosting: each new tree's contribution is scaled by $\nu \in (0, 1]$. Smaller values regularise the fit and generally improve generalisation at the cost of requiring more trees.

TENEXPCHAINFRISPRE FRED series identifier for the FRBSF's 10-year-ahead expected inflation measure derived from chain-type PCE price-index data; the sole dependent variable throughout this paper.

TIPS *Treasury Inflation-Protected Securities.* U.S. government bonds indexed to the CPI; by comparison with nominal Treasuries of the same maturity they reveal real yields and, via the BEIR, inflation expectations.

VC dimension *Vapnik–Chervonenkis dimension.* A combinatorial measure of hypothesis-class complexity; for the threshold classifier induced by the IF anomaly score, $d_{\mathrm{VC}} = 1$, yielding the tight bound of Theorem 5.1.

WCSS *Within-Cluster Sum of Squares.* The k-means objective $J = \sum_{j=1}^{k} \sum_{t:\, c_t = j} \|\boldsymbol{\phi}_t - \boldsymbol{\mu}_j\|^2$, shown in Proposition 6.1 to decrease monotonically under Lloyd's algorithm.

ZLB *Zero Lower Bound.* The practical floor on nominal short-term interest rates, constraining conventional monetary policy and tending to compress long-run inflation expectations.

The End